Pre-registration Protocol: Diagnostic Replication of GRACE (v1)

- Study ID: `grace_diag_replication_v1`
- Version: `prereg-v1`
- Registration Date: 2026-10-31 (Standardization update completing Step 4: 2026-11-02)
- Repository Version / Git Tag: `prereg-v1`
- Protocol Configuration File: `01_Author_Baseline_Unified/diagnostic_kit/configs/protocol.yaml` & `configs/protocol.yaml`
- Locked Thresholds File: `configs/thresholds.json` & `configs/thresholds_flat.json`
- Original Research Proposal Document: `00_Reports_and_Docs/PhuongAn2_De_cuong_nghien_cuu_GRACE_hop_nhat.pdf`

1. Test Exposure Disclosure & Exposure Ledger

The test sets of Devign and ReVeal were previously exposed during the initial exploratory phase (including 36 exploratory API test runs, the dev-100 subset, and preliminary H1–H3 motivation measurements).

To guarantee scientific validity and rigorous confirmatory standing:

1. All configuration choices, hyperparameters, token budgets, prompt templates, and 15 experimental arms are strictly locked on the official validation set prior to executing confirmatory test runs.
2. The optimal classification threshold τ^* for every arm is searched for and frozen entirely on the validation split (`configs/thresholds.json` and `runs/thresholds.json`), and is never tuned based on test results.
3. All exploratory results from earlier phases are treated strictly as academic motivation evidence and relegated to appendices, separated entirely from confirmatory results.
4. Any post-registration modification must be transparently documented with justification in `CHANGES_AFTER_PREREG.md`.

1.1. OSF Existing Data Declarations

According to Open Science Framework (OSF) registration classifications:

- Registration following analysis of the data: Applies to Tier 1 arms (B0, B1, B2) on the test sets of Devign and ReVeal because exploratory metrics (MCC, AUROC) were computed during the exploratory phase. In Table B of the paper, these contrast pairs are explicitly marked as "observed prior to registration".
- Registration prior to analysis of the data: Applies to all Tier 2–3 arms (B2-rand, B2-nolabel, B2-labelonly, B1-shuf, B1-full, B1-H2, B3, B3-nodiff, B3-swap, B3-flip) and all unexecuted confirmatory analyses: Table B (paired bootstrap contrasts with Holm correction), Table D (complementarity analysis, budget routing), and the 2-rater blind review audit.

1.2. Exposure Ledger & 3-Condition Audit for Keep vs. Rerun

In accordance with scientific advisory directives, the research team audited historical runs against three explicit conditions:

- Condition 1 (Protocol match): Cross-check the run manifest against `protocol.yaml`. All retained runs share checkpoint `Qwen3-Coder-30B-A3B-Instruct-Q4_K_M.gguf` (SHA-256 `f6050b4da22869aa3fb9d94bb3314e9fc5a027d4741c1f97761c387d6b2b2335`), backend `llama-cpp-python 0.3.16`, context limit 8,192 tokens, and canonical Figure 6 prompt ordering.
- Condition 2 (Logprob completeness and upper bounding): All records store logprobs for tokens `0` and `1` to compute $s = \text{sigmoid}(\text{logit}_1 - \text{logit}_0)$. For tokens absent from top-k (if any), a conservative upper bound $p \le p(\text{lowesttokenintop} - k)$ is applied alongside the `s_bounded` flag, ensuring a bounded rate $< 1\%$.

• Condition 3 (B2-rep cache isolation): Arm B2-rep has never run on GPU; when executed, `repeat: true` and `tag="repeat"` in diagnostic kit v1.1 are mandatory to isolate the cache, guaranteeing it never re-reads responses from the B2 cache.

Arm	Dataset	Split	First Run (UTC)	Commit	Metrics Viewed	Matches Frozen Protocol?	Action
B0	Devign	Val / Test	2026-11-01T04:21:38Z	`7b045e7`	MCC, AUROC, CM	Yes (`yes`)	Retain (`keep`)
B0	ReVeal	Val / Test	2026-11-01T04:21:38Z	`7b045e7`	MCC, AUROC, CM	Yes (`yes`)	Retain (`keep`)
B1	Devign	Val / Test	2026-11-01T10:17:36Z	`0785f78`	MCC, AUROC	Yes (`yes`)	Retain (`keep`)
B1	ReVeal	Val / Test	2026-11-01T10:17:36Z	`0785f78`	MCC, AUROC	Yes (`yes`)	Retain (`keep`)
B2	Devign	Val / Test	2026-10-31T17:05:34Z	`8f48ea9`	MCC, AUROC, CM	Yes (`yes`)	Retain (`keep`)
B2	ReVeal	Val / Test	2026-10-31T17:05:34Z	`8f48ea9`	MCC, AUROC, CM	Yes (`yes`)	Retain (`keep`)
B2-flip	Devign	Validation	2026-11-01T14:46:02Z	`484b901`	None (`none`)	Yes (`yes`)	Retain (`keep`)
B2-flip	Devign	Test	2026-11-01T14:46:02Z	`484b901`	Raw labels 230 samples	Yes (`yes`)	Rerun (`rerun`)
B2-flip	ReVeal	Validation	2026-11-01T14:46:02Z	`484b901`	None (`none`)	Yes (`yes`)	Retain (`keep`)
B2-flip	ReVeal	Test	Unexecuted	None	None (`none`)	Yes (`yes`)	Retain (`keep`)
B2-rep	Both	Val / Test	Unexecuted	None	None (`none`)	Yes (`yes`)	Fresh run without cache (`rerun_repeat_only`)
B2-rand	Both	Validation	2026-11-02T00:04:23Z	`484b901`	None (`none`)	Yes (`yes`)	Retain (`keep`)
Tier 2–3	Both	Val / Test	Unexecuted	None	None (`none`)	Yes (`yes`)	Pre-registered prior to analysis (`prior_to_analysis`)

2. Table of 11 Pre-registered Hypotheses (Table 4.3)

ID	Hypothesis Name	Specification & Decision Criteria (Pre-declared Practical Margin δ)
H1.1	GRACE does not outperform Zero-shot on Devign	Arm B2 (full GRACE) does not outperform B0 (zero-shot) in discriminative ability on Devign Test: Upper bound of 95% CI for $\Delta\text{MCC}(\text{B2} - \text{B0}) < 0.03$ and $\Delta\text{AUROC} < 0.02$ (practical margin $\delta = 0.02$).

ID	Hypothesis Name	Specification & Decision Criteria (Pre-declared Practical Margin δ)
H1.2	GRACE does not outperform Zero-shot on ReVeal & Fails to reach published baseline	Arm B2 does not outperform B0 on ReVeal Test: Upper bound of 95% CI for $\Delta\text{MCC}(\text{B2} - \text{B0}) < 0.03$ and $\Delta\text{AUROC} < 0.02$; GRACE fails to reproduce published performance in original paper: $\text{MCC}(\text{B2}) < 0.15$ on both datasets; $F_1(\text{B2})$ on Devign is lower than always-vulnerable baseline (62.95%).
H2.1	Demonstrations act via surface labels	In-context demonstrations operate primarily through surface label priming: Net label sensitivity ≥ 15 percentage points (proportion change when inverting labels $\text{B2} \leftrightarrow \text{B2} - \text{flip}$ minus natural variance $\text{B2} \leftrightarrow \text{B2} - \text{rep}$).
H2.2	Demo source code conveys no intrinsic signal	Demonstration source code adds no genuine signal: Upper bound of CI for $\Delta\text{AUROC}(\text{B2} - \text{B2} - \text{rand}) < 0.02$ and $\Delta\text{AUROC}(\text{B2} - \text{nolabel} - \text{B1}) < 0.02$; B2-labelonly recovers $\geq 80\%$ of the gap between B2 and B1.
H2.3	Contrastive pairs merely shift operating points	Contrastive demonstration pairing (B3) shifts operating decision points rather than increasing discriminative separation: B3 positive prediction rate is higher than B2 by ≥ 10 percentage points while upper bound of 95% CI for $\Delta\text{AUROC}(\text{B3} - \text{B2}) < 0.02$.
H3.1	LLMs do not read Code Property Graphs	Textualized CPG information provides no additional signal to the LLM: Upper bound of CI for $\Delta\text{AUROC}(\text{B1} - \text{B0}) < 0.02$ and $\Delta\text{AUROC}(\text{B1} - \text{B1} - \text{shuf}) < 0.02$; B1-H2 shows no significant difference under the same criterion.
H4.1	Threshold-free evaluation indicates near-random performance	When evaluated threshold-independently (AUROC), the entire LLM pipeline performs near random guess level: $\text{AUROC} \leq 0.62$ on both Devign and ReVeal across all LLM arms.
H5.1	Supervised baseline significantly outperforms LLM	Simple supervised classifier on frozen embeddings decisively outperforms LLM pipeline: $\Delta\text{MCC}(\text{SUP} - \text{LR} - \text{B2}) \geq 0.15$ with CI excluding 0; $\text{AUROC}(\text{SUP} - \text{LR}) \geq 0.68$.
H5.2	LLM provides no orthogonal post-query signal	LLM scores add no orthogonal signal after conditioning on 34 baseline features: Upper bound of 95% CI for $\Delta\text{MCC}(P_{\text{post}} - P_{\text{pre}}) < 0.02$ (identical 34 baseline features, identical StratifiedGroupKFold splits, identical threshold τ^*).
H5.3	Unpredictable expert deferral (Routing $q = 10\%$ Benefit vs Confidence & Complementarity)	At expert call budget $q = 10\%$, Ridge Benefit Router does not outperform Confidence Router: Paired Bootstrap CI for $\Delta\text{BalancedAccuracy}(\text{benefit} - \text{confidence})$ and ΔMCC have upper bounds $< \delta = 0.02$ (verdict `no_meaningful_benefit`); additionally $\Delta\text{BalancedAccuracy}(\text{benefit} - \text{random})$ CI upper bound < 0.01; LLM error repairs do not exceed the 95th percentile of an equal-positive-rate random expert (20 seeds).

ID	Hypothesis Name	Specification & Decision Criteria (Pre-declared Practical Margin δ)
H6.1	Retrieval mechanism misalignment	Retrieved demonstrations rarely match the underlying vulnerability mechanism; relevance does not imply correctness: "Neither" rate $\geq 40\%$ on Devign; accuracy difference between related and unrelated demos has CI containing 0.

Pre-specified Verdict Interpretation for All Pairwise Contrasts ($\Delta = \text{metric}(B) - \text{metric}(A)$)

- `benefit` (Evidence of benefit): CI lies strictly above 0 ($\text{lowerbound} > 0$) and $\Delta \geq \delta$.
- `no_meaningful_benefit` (No meaningful benefit): CI upper bound $< \delta$.
- `harm` (Evidence of harm): CI lies strictly below 0 ($\text{upperbound} < 0$).
- `inconclusive` (Inconclusive evidence): CI contains 0 or upper bound $\geq \delta$ while lower bound ≤ 0.

Pre-declared δ margins: General standard for practical equivalence margin is $\delta = 0.02$ (details: $\delta_{\text{MCC}} = 0.03$, $\delta_{\text{AUROC}} = 0.02$, $\delta_{\text{BA}} = 0.01$, $\delta_{\text{MCC_meta}} = 0.02$, $\delta_{\text{routing}} = 0.02$).

3. Analysis Rules & Statistical Methodology (Section 6)

1. Sanitization & Repository Freezing:

- Donor training data has all exact duplicate samples with validation and test sets removed (after normalizing whitespace and comments).

- In the donor bank, keep one representative for duplicate code groups with matching labels; eliminate code groups with conflicting labels.

- Near-duplicate pairs (Jaccard token 3-gram ≥ 0.9) are flagged for sensitivity analysis (reported on both full test and deduplicated test).

2. Locking 14 Optimal Decision Thresholds (τ^*):

- Arbitrary thresholds are strictly prohibited.

- Threshold optimization formula for dataset d, arm a, and seed r:

$$\tau^*(d, a, r) = \text{argmax}_{\tau \in \{0.01,\, 0.02,\, \ldots,\, 0.99\}} \text{MCC}(y_{\text{val}}(d), \mathbf{1}[s_{\text{val}}(d, a, r) \geq \tau])$$

- Tie-break rule: In case of ties, select the threshold closest to 0.5 (`closest_to_0.5`); if tied in distance to 0.5, take the smaller threshold. Use only valid validation samples (`valid == True`); thresholds yielding undefined MCC (all predictions in one class) are discarded.

- Threshold Inheritance Rule: Label intervention and repetition probes ($B2{-}\text{rep}$, $B2{-}\text{flip}$, $B3{-}\text{flip}$, $B3{-}\text{swap}$) must not select τ^* independently on validation, but instead directly inherit the locked τ^* of the corresponding base arm ($B2$ or $B3$). Re-optimizing thresholds for an arm with inverted labels or repetitions would "repair" the intervention effect and confound threshold tuning variance with the experimental treatment.

- All 14 optimal thresholds τ^* are locked into `configs/thresholds.json` and `runs/thresholds.json` prior to any test inference:

- Devign LLM Arms: `devign|B0|42` = 0.08, `devign|B1|42` = 0.88, `devign|B2|42` = 0.70, `devign|B2-flip|42` = 0.70 (inherited from B2), `devign|B2-rep|42` = 0.70 (inherited from B2).

- ReVeal LLM Arms: `reveal|B0|42` = 0.44, `reveal|B1|42` = 0.74, `reveal|B2|42` = 0.12, `reveal|B2-flip|42` = 0.12 (inherited from B2), `reveal|B2-rep|42` = 0.12 (inherited from B2).

- Meta-Models: `devign|P-PRE|42` = 0.54, `devign|P-POST|42` = 0.60; `reveal|P-PRE|42` = 0.38, `reveal|P-POST|42` = 0.10.

3. Confirmatory Statistics & Pairwise Contrasts (Table B):

- Paired Bootstrap $B = 5000$ iterations with fixed seed `20261117`.

- Cluster Units:

- Devign: Cluster Bootstrap by `commit_id` (2,068 independent clusters on Test $N = 2,732$).

- ReVeal: Cluster Bootstrap by File-level Hashes (`file_hash`, 646 independent clusters on Test $N = 2,274$, completely resolving the chrome/debian cluster collapse risk identified in Step 0).

- Meta-model fold splitting: `StratifiedGroupKFold` (3 folds) on Validation ensuring independence of code groups.

- Family-wise error rate control: Holm-Bonferroni method (`family_correction: holm`).

4. Complementarity & Routing Analysis:

- 34 baseline features P_{pre} include 5 foundation model logits, ensemble probability p, $|p - 0.5|$, MLP-Lexical discrepancy, and 26 code metrics.

- 4 LLM features include: continuous score s, `valid` flag of candidate score, interaction $s \cdot p$, and discrepancy $s - p$. When no valid score is available, the meta-model uses symmetric fallback ($s = 0.5, valid = 0$); raw greedy `response_valid` is reported separately.

- Meta-model employs Logistic Regression ($C = 1.0$), 3-fold Out-Of-Fold (OOF).

- Routing evaluation at 5%, 10%, 25% expert call budgets with direct Paired Bootstrap CI between Benefit Router vs. Confidence Router, 20 random seeds for RANDOM baseline, and Oracle ceiling.

4. 15 Experimental Arms Matrix & Probe Construction Rules (Section 6.3)

Tier	Arm ID	Prompt Content & Probe Specification	Difference Relative to Base Arm	Corresponding RQ
Floor & Anchors	ALL-0, ALL-1, RAND	Predict constant 0, constant 1, or random sampling matching label prior	No LLM call; label distribution baseline	All
Floor & Anchors	SUP-LR, SUP-MLP, SUP-METRIC	Logistic Regression on CodeT5 768d, MLP 768→128→1, LR on 26 code metrics	Supervised anchor on frozen embeddings (3 seeds: 42, 43, 44)	RQ5
Tier 1	B0	Role + domain + target code + binary question	Zero-shot; no CPG graph, no demonstration	RQ1
Tier 1	B1	B0 + CPG nodes & edges (prefix 40 nodes / 80 edges)	Adds textualized CPG graph as reimplemented	RQ1, RQ3
Tier 1	B2	B1 + 1 demonstration retrieved from clean donor bank with true label	Full GRACE pipeline	RQ1, RQ2
Tier 1	B2-flip	B2 with inverted demonstration label ($1 \leftrightarrow 0$) in prompt	Measures net label sensitivity	RQ2
Tier 1	B2-rep	B2 repeated with identical prompt (bypassing cache)	Measures natural variance from batching / GPU nondeterminism	RQ2
Tier 2	B2-rand	B1 + 1 random demonstration from clean donor bank with true label	Demonstration code unrelated to mechanism (3 seeds: 42, 43, 44)	RQ2
Tier 2	B2-nolabel	B1 + retrieved demonstration; prompt altered to "The following is a related example." (unlabelled)	Strips demonstration label	RQ2

Tier	Arm ID	Prompt Content & Probe Specification	Difference Relative to Base Arm	Corresponding RQ
Tier 2	B2-labelonly	B1 + prompt "A function similar to the one above was labelled Vulnerable/ Non-vulnerable." (no code)	Surface label priming prompt only	RQ2
Tier 2	B1-shuf	B0 + CPG from an unrelated function with matched node count ($\pm 10\%$), same budget	Graph conveys no target semantics (3 seeds: 42, 43, 44)	RQ3
Tier 3	B1-full	B0 + CPG expanding up to full 1,500 token budget	Altered graph truncation policy (larger node prefix)	RQ3
Tier 3	B1-H2	B0 + H2 guarded execution evidence cluster packed to 1,500 tokens	Packages execution trace paths instead of raw prefix	RQ3
Tier 3	B3	B1 + contrastive demo pair (1 vuln, 1 safe) + Key difference block	Proposed Contrastive ICL method	RQ2, RQ4
Tier 3	B3-nodiff	B3 with Key difference block omitted	Isolates effect of differential explanation block	RQ2
Tier 3	B3-swap	B3 with order of the two contrastive demonstrations swapped	Measures recency / order bias	RQ2
Tier 3	B3-flip	B3 with labels of both demonstrations inverted ($1 \leftrightarrow 0$)	Measures label sensitivity under balanced demonstration design	RQ2

5. Prompt Contract, Token Budget & Section Ordering (Figure 6 & Section 6.4)

1. Canonical Section Ordering (Figure 6 & Author Code):

`[role_domain, target_code, question, node_info, edge_info, demonstration, answer]`

Invariant rule: Target code comes first, graph in the middle, demonstration at the end immediately preceding the answer prompt. Target labels never appear in the prompt.

2. Maximum Token Budget (8,192 Token Context):

- Target source code: $\leq 3,000$ tokens.

- Demonstration: $\leq 1,500$ tokens per demo.

- CPG Graph: $\leq 1,500$ tokens.

- Reserve: 64 tokens.

- Measurement unit: Strictly measured via the model's tokenizer (`Qwen3-Coder-30B-A3B-Instruct`).

3. Output, Score & Validity for Executed Inference:

- The Kaggle notebook invokes `llama-cpp-python` directly, reading all logits at the first token position after the prompt; resolves separate token IDs for `0` and `1`, then computes `logprob_0`, `logprob_1`. Continuous score $s = \frac{p(1)}{p(0)+p(1)} = \text{sigmoid}(\text{logit}_1 - \text{logit}_0)$. The `top5` dictionary is preserved for diagnostics, not as a prerequisite for score extraction.

- `valid` is the score validity flag: both token IDs must be resolvable, distinct, within vocabulary; logits must be finite and s computable. If logits are missing, non-finite, tokenizer mismatched, or backend fails, no valid score exists; score is not

forced to 0. In the current notebook, pre-record errors abort the run or checkpoint completed records, rather than creating artificial `valid=false` records.

- `raw` is the greedy token over the entire vocabulary; `response_valid = true` strictly when decoded raw is `'0'` or `'1'`. Raw differing from both labels can occur even when `valid=true` and s is finite. Separately report counts/rates of `valid` and `response_valid`; finite scores are not discarded merely because raw is non-compliant. `pred_argmax` of the score uses $s \geq 0.5$; when evaluated against locked thresholds, $s \geq \tau^*$ is used, without confounding these predictions with raw greedy output.

- The legacy OpenAI-compatible parser in `llm_backend.py` used `top_logprobs` requiring both tokens `0` and `1` in top-k. That rule pertained to legacy API mode, not the Kaggle notebook's full logit scoring.

6. Stopping Rules

1. SUP-LR Advantage Disappears: If after deduplication and label auditing SUP-LR no longer significantly outperforms B2, halt the study to review preprocessing before drafting.
2. Prompt Audit Failure: If empty demonstration rate $> 1\%$, missing graph rate $> 5\%$ (excluding flagged Joern errors), or target label leakage > 0, halt pipeline, correct prompt contract, and rerun all affected arms.
3. Backend Smoke Test Failure: If valid response rate containing token 0/1 in top-5 logprob $< 99\%$, do not execute confirmatory matrix until backend achieves quantitative stability. This is Step 1 API smoke test criteria, not the validity definition in Kaggle full-logit inference.
4. Strict Protocol Freeze: Absolutely no new arms, no additional models, and no threshold tuning after observing test data. Any post-hoc analyses must be labeled "exploratory post-hoc" in the final manuscript.

7. Frozen Hashes, Manifests & Provenance Tables

7.1. SHA-256 Checksum Table for Data & Manifests

Dataset	Partition / Asset	Rows (Samples)	Positive Rate (Vuln)	File Path / Permanent Link	SHA-256 Hash
Devign	Raw Train	21,854	45.84%	`data/raw/devign_train.jsonl`	`e456e20da10aa9a30df406d78dcf5f15f91da8c49fd5028584ca248b03f13f35`
Devign	Raw Validation	2,732	43.45%	`data/raw/devign_validation.jsonl`	`7318bc4aab8fecdac3ce1afe51c7833714be1b807c5437ab2c7dc0384a6543e0`
Devign	Raw Test	2,732	45.94%	`data/raw/devign_test.jsonl`	`7f77094d9328624036cbd43ab1281aaf0c3378e220d5b3a10ba0a98922ab39c6`
Devign	Clean Donors Bank	21,691	45.91%	`data/manifests/devign_donor_clean.jsonl`	`5eda46695ea98091168c2a87591010b4fe6c08597a376c3361b4cf40297d1dfe`
Devign	Manifest Concatenation	-	-	`data/manifests/devign_manifest_stats.json`	`5a084a10e5c191039e56e351bf8d722fe017424f159a84d6b92be86a1d0f4952`

Dataset	Partition / Asset	Rows (Samples)	Positive Rate (Vuln)	File Path / Permanent Link	SHA-256 Hash
Devign	Val Subset 600 (Seed 42)	600	43.50%	`data/specs/devign_val_subset600.jsonl`	`88f749c000a7b110ed87641d9d5dc4bc912b9f2fdb3dcc2a97753238a4dbcdfd`
Devign	Test Specs	2,732	45.94%	`data/specs/devign_test.jsonl`	`bd0f4d025afd051c064ab58c1495a887c5851b6408e2d47d39ae64d421b898c6`
ReVeal	Raw Train	18,187	9.90%	`data/raw/reveal_train.jsonl`	`87c6ec979d8724d1db25217ae6c5637fedaca5350fc954baab4c615efc56f66c`
ReVeal	Raw Validation	2,273	9.24%	`data/raw/reveal_validation.jsonl`	`03ccd535ecc5c85601377d1880f05307d20577e195cc8808d2c5cbd880725d50`
ReVeal	Raw Test	2,274	10.11%	`data/raw/reveal_test.jsonl`	`5132c4a2cab7cea81435b5039af964cab184e6ce7b4d9e2df524962b0f6d45f3`
ReVeal	Clean Donors Bank	17,636	9.57%	`data/manifests/reveal_donor_clean.jsonl`	`9e891cf5b8dfa93c2c3d9d5039e328c5d36c115d173051c5991fc0939e1d372d`
ReVeal	Manifest Concatenation	-	-	`data/manifests/reveal_manifest_stats.json`	`59498c4a4ab038babe558675c183c2b104b0145590fb0accf8b02cd5440d57f7`
ReVeal	Val Subset 600 (Seed 42)	600	9.17%	`data/specs/reveal_val_subset600.jsonl`	`72ce24b2629588a7edf5206c99b26865b630a489580a6b03964c368fad79858e`
ReVeal	Test Specs	2,274	10.11%	`data/specs/reveal_test.jsonl`	`26d79cec7a31e45a99cf8f22266aa793f323eacb6328cf8d5bd2fc1b34131deb`

Artifact Reconciliation Note: Display names in the path column follow the preregistration protocol; GitHub links resolve to frozen raw/stats artifacts under snapshot `research_freeze_20261101`, while the two Google Drive links point to `donors.jsonl` files from the same snapshot (Devign SHA-256 `7727d2793fbb9f305e547bd6e7126ae70f2e0ee97e2b7c5c260322c6b6af5a62`; ReVeal SHA-256 `6bf44ab86c8524d239e610fdccff5916cfc402fef7945d7cbe7f695e287abbce`). SHA-256 values printed in the table reflect frozen manifest checkpoints. The two standardized 600-sample validation subsets `data/specs/devign_val_subset600.jsonl` (SHA-256 `88f749c000a7b110ed87641d9d5dc4bc912b9f2fdb3dcc2a97753238a4dbcdfd`) and `data/specs/reveal_val_subset600.jsonl` (SHA-256 `72ce24b2629588a7edf5206c99b26865b630a489580a6b03964c368fad79858e`) match 100% exactly with the table above and have explicit `.gitignore` inclusion exceptions.

7.2. Model Checkpoint & Inference Environment (Standardized Provenance)

• Architecture and tokenizer: `Qwen/Qwen3-Coder-30B-A3B-Instruct`; tokenizer at revision `b2cff646eb4bb1d68355c01b18ae02e7cf42d120`.

• Model weights: GGUF from `ggml-org/Qwen3-Coder-30B-A3B-Instruct-GGUF`, file `Qwen3-Coder-30B-A3B-Instruct-Q4_K_M.gguf`, quantization `Q4_K_M`, SHA-256 `f6050b4da22869aa3fb9d94bb3314e9fc5a027d4741c1f97761c387d6b2b2335`. Manifests of executed B0, B1, B2, B2-flip, and B2-rand runs record this exact identifier.

• Inference backend: `llama-cpp-python 0.3.16` invoking `Llama(...)` directly on Kaggle GPU 2× Tesla T4, with `n_ctx=8192`, `n_batch=256`, `n_gpu_layers=-1`, `logits_all=True`; scoring from first token logits of labels `0` and `1`. This matches the configuration in notebooks and run manifests, distinct from REST/Cloudflare `llama-server`.

• Kaggle model source: Notebook B2 resolves the GGUF file in output of `quangdung1111/qwen3coder30b-prepare-notebook`; other arm notebooks attach the GGUF file as input and verify SHA-256 prior to inference. Slug `tnthnhneee/qwen3-30b-a3b/gguf/coder-instruct-ud-q4_k_xl` in old protocol drafts does not identify the executed `Q4_K_M` file.

• Fallback model scope: None of the reconciled inference runs B0/B1/B2/B2-flip/B2-rand used vLLM or Qwen2.5-Coder. Those options in `protocol.yaml` were prospective fallbacks and are not combined with GGUF results.

• Note on locked files: Both `protocol.yaml` files are preserved as historical artifacts because `thresholds.json` stores the locked protocol SHA-256. This section represents post-hoc reconciliation against notebooks and outputs without altering the timing or contents of the locked protocol.

8. Pre-registration Acceptance Checklist (Appendix C - Proposal)

• [x] Step 0: Historical metrics reproduced from sample-level files (discrepancy ≤ 0.001); repair-harm table includes ALL-0 / ALL-1 / RANDOM rows; paired bootstrap 5,000 reps; cluster unit resolved: Devign (2,068 commit groups), ReVeal (646 file hash groups); StratifiedGroupKFold OOF; Paired Bootstrap CI for Routing q=10% (Benefit vs Confidence).

• [x] Backend Smoke Test: $\geq 99\%$ prompts contain both tokens 0 and 1 in top-k ($k = 20$); repeated call agreement $\geq 98\%$ (100% observed); throughput and latency measured; fail-fast 0 retry on context limit exceedance.

• [x] Data Manifests: SHA-256 hashes frozen; single deduplication audit completed (excluded 18 Devign cases, 129 ReVeal cases); donor bank sanitized (21,691 Devign, 17,636 ReVeal).

• [x] Testing & Prompt Audit: pytest tests/ green (100% pass); prompt audit: label leakage $= 0$, empty demos $\leq 1\%$ (0.0% on Devign, 0.7% on ReVeal); missing graphs flagged; 30 prompts per dataset curated in `manual_sample.jsonl`.

• [x] Research Protocol: `protocol.yaml` completed; `preregistration.md` finalized, tagged `prereg-v1`, and ready for OSF submission prior to confirmatory test inference.

• [x] Threshold Lock Step 5: τ^* of every arm selected on validation via `grace_diag.thresholds`, locked (`locked_sha256`) in `thresholds.json` and committed prior to test runs; repeated/inverted probes ($B2{-}\text{flip}$, $B2{-}\text{rep}$, $B3{-}\text{flip}$, $B3{-}\text{swap}$) inherit base arm τ^*.

• [x] Test Exposure Ledger: Populated for all arms executed prior to pre-registration (`exposure_ledger`); retained arms have `manifest.json` matching frozen protocol and records with logprobs sufficient to reconstruct s; zero B2-rep runs ever read from cache (`repeat: true`, `tag="repeat"`).

• [x] Execution Schema B.2: All test runs produce `records.jsonl` matching schema B.2 and environment `manifest.json`; caching keyed by complete execution identity; candidate-score validity and raw verdict validity separated transparently.

• [x] Meta-Models & Routing Step 6: SUP-LR/SUP-MLP/SUP-METRIC evaluated across 3 seeds; P_pre/P_post share identical StratifiedGroupKFold folds with symmetric fallback; routing evaluated with 20 random seeds, oracle ceiling, and paired bootstrap CI at q=10%.

• [x] Final Evaluation & Tables A/B/C/D: Tables A, B, C, D and both figures generated from a single unified script (`evaluate_final.py`), without manual editing; Table B includes pre-declared decision verdicts under $\delta = 0.02$

and degenerate bootstrap count tracking.

• [x] 11 Scientific Hypotheses Cross-Check: Hypothesis mapping table 4.3 (H1.1–H6.1) locked with pre-declared support / reject / inconclusive criteria based on practical margin δ prior to test execution.

• [ ] Blind Review & Label Audit (Step 8): Blind review κ (retrieval and 60-case packets), conditional analysis tables; label audit on 100 cases per dataset (plan and packet formats finalized, executed in parallel with Step 5 on GPU).

• [x] Independent OSF Reproducibility Bundle: Reproducible pack runnable from README on a clean machine; offline portion requires no GPU; no API keys in logs; 39 artifacts bundled into ZIP with verified SHA-256 cross-check.